

Pantoprazole for Cardiovascular Diseases

Novell Pharmaceutical Laboratories

Penyakit Penyakit Cardiovascular

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graph TD; A[Penyakit Penyakit Cardiovascular] --> B[Coronary Artery Disease (CAD)/Penyakit Jantung Koroner (PJK)]; A --> C[Atrial Fibrillation (AF)]; A --> D[Venous Thromboembolism (VTE)/Tromboemboli Vena]; A --> E[Penyakit Katup Jantung Mekanik (Pasca-Valve Replacement)]; A --> F[Peripheral Artery Disease (PAD)]; G[Ischemic Stroke (Stroke Iskemik)]
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Coronary Artery
Disease
(CAD)/Penyakit
Jantung Koroner
(PJK)

Atrial Fibrillation (AF)

Venous
Thromboembolism
(VTE)/Tromboemboli
Vena

Penyakit Katup
Jantung Mekanik
(Pasca-Valve
Replacement)

Peripheral Artery
Disease (PAD)

Ischemic Stroke (Stroke Iskemik)

Coronary Artery Disease (CAD)

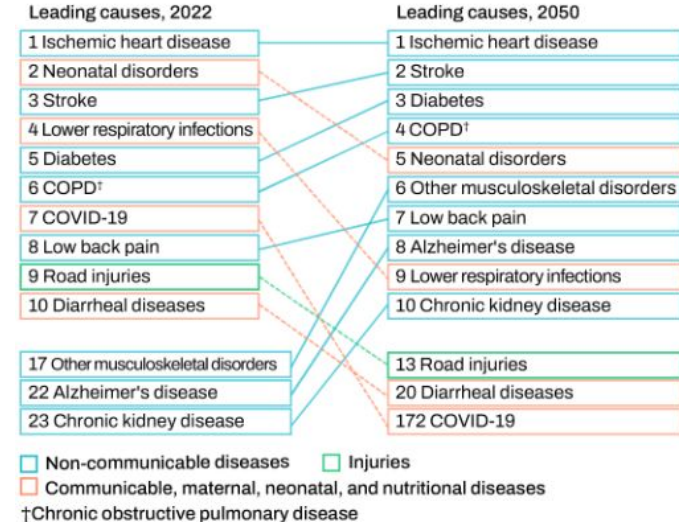
1st

Secara global, CAD adalah penyebab kematian nomor satu

197
Juta

Prevalensi global berdasarkan Global Burden of Disease (GBD) 2019, sekitar 197 juta kasus penyakit jantung iskemik

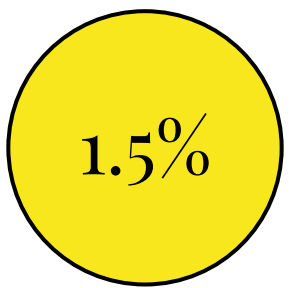
Leading causes of disease burden* worldwide in 2022 versus 2050



*Measured in number of disability-adjusted life years (DALYs)



**Institute for Health
Metrics and Evaluation**



Riskesdas 2018 melaporkan prevalensi PJK (diagnosis dokter) sebesar 1,5% populasi



Studi GBD 2019 untuk Indonesia mengestimasi 3,07 juta kasus IHD (PJK) di Indonesia tahun 2019

Coronary Artery Disease (CAD)

Journal of Epidemiology and Global Health (2024) 14:193–212
<https://doi.org/10.1007/s44197-024-00187-8>

RESEARCH ARTICLE



The 30 Years of Shifting in The Indonesian Cardiovascular Burden—Analysis of The Global Burden of Disease Study

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3.1.2 Prevalence and incidence

All-age CVD prevalence increased by 120%, from 6968 thousand (95% UI: 5919–8174) in 1990 to 15,348 thousand (95% UI: 13,000–18,080) in 2019. Age-standardized prevalence rates (ASPR) increased by 6.1%, from 6781/100,000 (95% UI: 5749–7978) in 1990 to 7196.3/100,000 (95% UI: 6089.6–8495.3) in 2019. Based on sub-categories, stroke and PAD I had the highest prevalence rates, comprising 4918 thousand (95% UI: 4417–5484) with an ASPR of 2097/100,000 and 3325 (95% UI: 2850–3803) with an ASPR of 1640/100,000, respectively. This was followed by IHD, which had a prevalence rate of 3066 thousand (95% UI: 2621–3621) with an ASPR of 1495/100,000. All CVD



GBD

Ischemic Stroke



101
Juta

Prevalensi global semua stroke tahun 2019 mencapai 101 juta kasus (survivor stroke)

1,09
%

Risesdas 2018 menunjukkan prevalensi stroke (segala jenis) 10,9 per mil (1,09%)

62%

Sekitar 62% di antaranya adalah stroke iskemik ➤ >60 juta orang yang pernah mengalami stroke iskemik di dunia

4,9
Juta

GBD 2019 melaporkan prevalensi 4,918 juta kasus stroke di Indonesia

Atrial Fibrillation (AF)

AF adalah aritmia
jantung tersering

Prevalensi global AF tahun 2010 diperkirakan 33,5 juta kasus

59,7

GBD secara global 2019 melaporkan 59,7 juta kasus AF/flutter

1,53
Juta

GBD 2019 mengestimasikan 1,53 juta kasus AF/flutter di
Indonesia tahun 2019

Venous Thromboembolism (VTE)

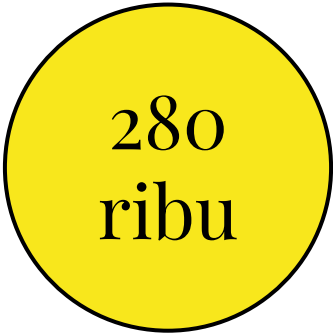
Deep Vein Thrombosis
(DVT)

Pulmonary Embolism (PE)

59,7

Insidens global VTE diperkirakan 10 juta kasus VTE terjadi
setiap tahun di seluruh dunia

Katup Jantung Mekanik (Pasca-Valve Replacement)



280
ribu

Diperkirakan ± 280.000 operasi penggantian katup dilakukan di dunia per tahun, sekitar setengahnya menggunakan katup mekanik



2-3
Juta

Akumulatif sejak era 1960-an, telah terpasang $>2-3$ juta katup mekanik di seluruh dunia

Di Indonesia, beban penyakit katup (terutama akibat rheumatic heart disease) masih signifikan, namun data spesifik pasien dengan katup mekanik belum terdokumentasi luas

Peripheral Artery Disease (PAD)



236
Juta

Analisis 2015 menunjukkan 236 juta orang usia ≥ 25 tahun hidup dengan PAD di seluruh dunia



3,3
Juta

Di Indonesia, GBD 2019 mencatat prevalensi 3,325 juta kasus

PAD cenderung meningkat seiring prevalensi diabetes dan merokok yang tinggi. Riskesdas belum secara spesifik mensurvei PAD, namun data GBD menunjukkan PAD menyumbang 9,2% dari total beban penyakit kardiovaskular di Indonesia

PPI untuk Penyakit Kardiovaskular



PEDOMAN TATA LAKSANA SINDROM KORONER AKUT

Edisi ke-5

Perhimpunan Dokter Spesialis
Kardiovaskular Indonesia (PERKI)
Jakarta
2024

Penghambat pompa proton mengurangi risiko perdarahan saluran cerna bagian atas pada pasien yang mendapatkan anti-platelet.

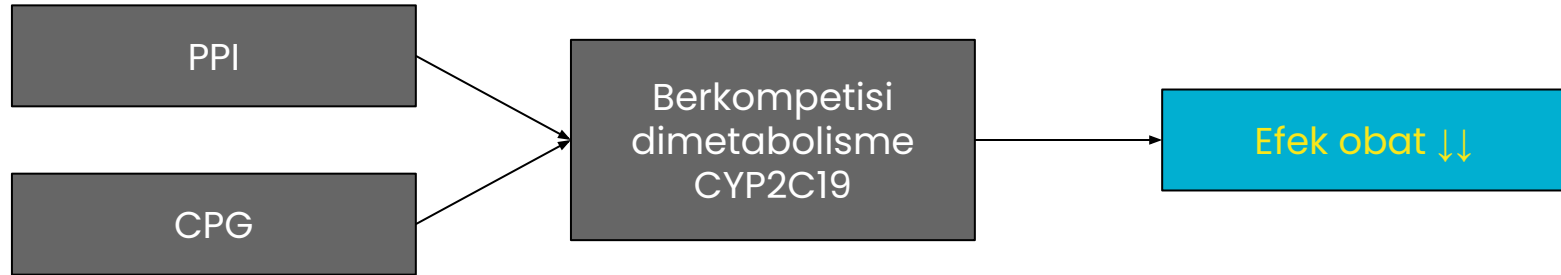
Terapi PPI diindikasikan untuk pasien yang menerima regimen anti-trombotik apa pun, yang berisiko tinggi mengalami perdarahan saluran cerna

“PPI yang menghambat CYP2C19, khususnya omeprazole dan esomeprazole, dapat mengurangi respons farmakodinamik terhadap clopidogrel”

Pantoprazole Untuk Obat Kardiovaskular

Penyakit seperti CAD, stroke iskemik, AF, VTE, pasca operasi katup mekanik maupun PCI menggunakan CPG sebagai terapi medika mentosa

PPI yang dikonsumsi bersamaan dengan clopidogrel (CPG) akan bersaing secara kompetitif untuk dikelola oleh enzim (Metabolisme minimal lewat CYP2C19 → interaksi obat (misal dengan klopido~~g~~rel) rendah) yang sama



Pantoprazole merupakan PPI yang relatif aman untuk obat clopidogrel

Wedemeyer RS et al 2014 Drug Interaction Pantoprazole

Pantoprazole memiliki potensi interaksi obat yang lebih rendah dengan PPI lain

Concomitant drug	Effect of PPI on concomitant drug				
	Esomeprazole	Lansoprazole	Omeprazole	Pantoprazole-Na	Rabeprazole
Antacid	Unknown	Conflicting results [108, 111]	None [64]	None [83]	None [123]
Phenazone (antipyrene)	Unknown	+ Clearance [106]	+ Clearance [125]	None [84]	Unknown
Bortezomib	Unknown	Unknown	None [42]	Unknown	Unknown
Caffeine	Unknown	None [126]	Conflicting results [126, 127]	None [85, 126]	Unknown
Carbamazepine	Unknown	Unknown	+ Clearance [128]	None [86]	Unknown
Oral contraceptives	Unknown	Conflicting results [129]	Unknown [125]	None [98]	Unknown
Ciclosporin	Unknown	Unknown	Conflicting results [70, 71, 130]	None [88]	Unknown
Cinacalcet	Unknown	Unknown	Unknown	None [87]	Unknown
Ciprofloxacin ER	Unknown	Unknown	None [43]	Unknown	Unknown
Citalopram	Unknown	Unknown	+ Clearance ^a [44]	Unknown	Unknown
Clarithromycin	Unknown	Unknown	None [45]	None [45]	Unknown
Clopidogrel	+ Absorption [39]	None [39]	+ Absorption [37]	None [37]	Unknown
Diazepam	+ Clearance [80–82, 131]	None [107]	+ Clearance [53, 54]	None [82, 89]	None ^b [55]
Diclofenac	Unknown	Unknown	None [65]	None [90]	Unknown
Digoxin	Unknown	Unknown	+ Absorption [132]	None ^c [91]	+ Absorption [133]
Ethanol	Unknown	None [134]	None [134]	None [92]	Unknown
Etravirine	Unknown	Unknown	+ Clearance [46]	Unknown	Unknown
Gemifloxacin	Unknown	Unknown	None [47]	Unknown	Unknown
Glibenclamide	Unknown	Unknown	Unknown	None [93]	Unknown
Ivabradine	Unknown	None [48]	None [48]	Unknown	Unknown
Levothyroxine	Unknown	Unknown	Unknown	None [94]	Unknown
Metoprolol	Unknown	Unknown	None [66]	None [95]	Unknown
Naproxen	Unknown	Unknown	None [65]	None [96]	Unknown
Nifedipine	Unknown	Unknown	+ Absorption + Clearance [67]	None ^d [97]	Unknown
Phenprocoumon	Unknown	Unknown	+ Clearance [63]	None [99]	Unknown
Phenytoin	+ Clearance [80, 131]	None [110]	+ Clearance [54, 58, 135]	None [100]	None [120]
Piroxicam	Unknown	Unknown	None [65]	None [101]	Unknown
Tacrolimus	Unknown	+ Clearance [117]	Unknown	None [102]	None [117, 122, 136]
Theophylline	Unknown	Conflicting results [113, 114]	None [68, 113]	None [103, 113]	None [121]
Warfarin	+ Clearance ^e [80, 131]	None [111]	+ Clearance ^e [59–61]	None [104]	None [121]

Pharmacodynamic Evaluation of Pantoprazole Therapy on Clopidogrel Effects

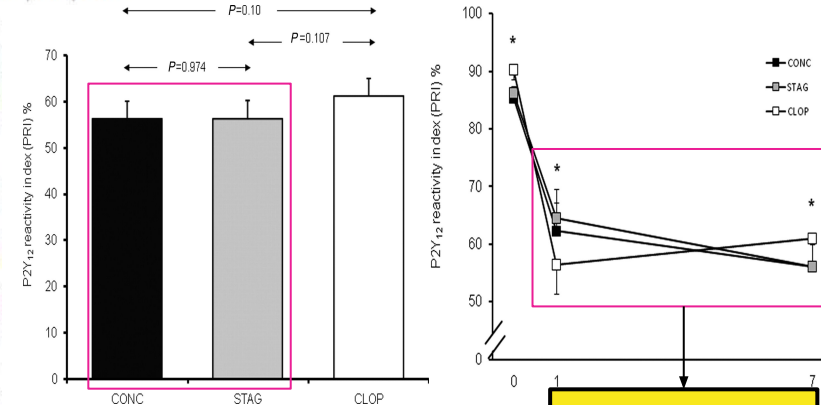
Results of a Prospective, Randomized, Crossover Study

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Naveen Secheran, MD; Murali K. Kodali, MD; Andrew Darlington, DO;
John-Paul Pham, MD; Antonio Tello-Montoliu, MD, PhD; Ronald K. Charlton, PhD;
Theodore A. Bass, MD; Dominick J. Angiolillo, MD, PhD

Background—Safety concerns have recently emerged based on a drug interaction between clopidogrel and proton pump inhibitors leading to reduced pharmacodynamic effects. However, whether such drug interaction is a class effect or a drug effect and if this can be modulated by timing of drug administration remains a matter of debate. The aim of this study was to assess the impact of high-dose pantoprazole therapy, a proton pump inhibitor with low potential to interfere with clopidogrel metabolism, administered concomitantly or staggered, on clopidogrel-mediated pharmacodynamic effects.

Methods and Results—This was a prospective, randomized, crossover study conducted in 20 healthy volunteers. Subjects were randomly assigned to receive pantoprazole (80 mg daily) administered concomitantly (CONC) or staggered by 8 to 12 hours (STAG) for 1 week on a background of clopidogrel therapy (600-mg loading dose followed by a 75-mg maintenance dose during all phases) in a crossover fashion with a 2- to 4-week washout period between treatments. All subjects had a 1-week treatment phase with a clopidogrel-only regimen with a 2- to 4-week washout period from randomization sequence. Platelet function was assessed by flow cytometric analysis of the status of phosphorylation of the vasodilator-stimulated phosphoprotein, light transmittance aggregometry after adenosine diphosphate stimuli, and VerifyNow P2Y₁₂ system at 3 time points: baseline, 24 hours after loading dose, and 1 week after maintenance dose. The primary end point was the comparison of P2Y₁₂ reactivity index assessed by vasodilator-stimulated phosphoprotein at 1 week. After 1 week, there were no significant difference in P2Y₁₂ reactivity index between the CONC and STAG regimens (least-squares mean±SEM, 56.0±3.9% versus 56.1±3.9%; $P=0.974$), as well as when compared with the clopidogrel-only regimen (61.0±3.9%; $P=0.100$ versus CONC and $P=0.107$ versus STAG). Further, no differences were observed at baseline and 24 hours between regimens. Concordant results were obtained by light transmittance aggregometry and VerifyNow P2Y₁₂ assays.

Conclusions—Pantoprazole therapy used at high doses is not associated with modulation of the pharmacodynamic effects of clopidogrel, irrespective of timing of drug administration.



Efek CPG pada pengguna pantoprazole tidak mengalami penurunan

Setelah 1 minggu pengobatan, tidak ada perbedaan signifikan dalam efektivitas clopidogrel

Penggunaan pantoprazole dalam dosis tinggi aman dan tidak mengurangi efektivitas clopidogrel dalam mencegah pembekuan darah

Safety of Proton Pump Inhibitors Based on a Large, Multi-Year, Randomized Trial of Patients Receiving Rivaroxaban or Aspirin

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See Covering the Cover synopsis on 587; see editorial on page 604.

BACKGROUND & AIMS: Proton pump inhibitors (PPIs) are effective at treating acid-related disorders. These drugs are well tolerated in the short term, but long-term treatment was associated with adverse events in observational studies. We aimed to confirm these findings in an adequately powered randomized trial. **METHODS:** We performed a 3 × 2 partial factorial double-blind trial of 17,598 participants with stable cardiovascular disease and peripheral artery disease randomly assigned to groups given pantoprazole (40 mg daily, n = 8791) or placebo (n = 8807). Participants were also randomly assigned to groups that received rivaroxaban (2.5 mg twice daily) with aspirin (100 mg once daily), rivaroxaban (5 mg twice daily), or aspirin (100 mg) alone. We collected data on development of pneumonia, *Clostridium difficile* infection, other enteric infections, fractures,

gastric atrophy, chronic kidney disease, diabetes, chronic obstructive lung disease, dementia, cardiovascular disease, cancer, hospitalizations, and all-cause mortality every 6 months. Patients were followed up for a median of 3.01 years, with 53,152 patient-years of follow-up. **RESULTS:** There was no statistically significant difference between the pantoprazole and placebo groups in safety events except for enteric infections (1.4% vs 1.0% in the placebo group; odds ratio, 1.33; 95% confidence interval, 1.01–1.75). For all other safety outcomes, proportions were similar between groups except for *C. difficile* infection, which was approximately twice as common in the pantoprazole vs the placebo group, although there were only 13 events, so this difference was not statistically significant. **CONCLUSIONS:** In a large placebo-controlled randomized trial, we found that pantoprazole is not associated with any adverse event when used for 3 years, with the possible exception of an increased risk of enteric infections. *ClinicalTrials.gov* Number: NCT01776424.



Table 2. Cardiovascular Events, Cancers and Hospitalizations

Outcome	Pantoprazole, 40 mg od (n = 8791)		Placebo (n = 8807)		Pantoprazole vs placebo HR (95% CI)	P value
	First events, n (%)	Annual rate, %/y	First events, n (%)	Annual rate, %/y		
Primary efficacy outcome*						
MI, stroke, or cardiovascular death	691 (7.9)	2.66	668 (7.6)	2.57	1.04 (0.93–1.15)	.51
Secondary efficacy outcomes*						
MI, ischemic stroke, CHD death, or ALI	588 (6.7)	2.27	572 (6.5)	2.20	1.03 (0.92–1.16)	.61
MI, ischemic stroke, cardiovascular death, or ALI	707 (8.0)	2.72	683 (7.8)	2.63	1.04 (0.94–1.15)	.50
Death						
All cause	630 (7.2)	2.37	614 (7.0)	2.31	1.03 (0.92–1.15)	.63
Cardiovascular	343 (3.9)	1.29	333 (3.8)	1.25	1.03 (0.89–1.20)	.69
Non-cardiovascular	287 (3.3)	1.08	281 (3.2)	1.06	1.02 (0.87–1.21)	.78
CHD	194 (2.2)	0.73	200 (2.3)	0.75	0.97 (0.80–1.18)	.94
Individual efficacy outcomes						
MI	252 (2.9)	0.96	267 (3.0)	1.02	0.94 (0.79–1.12)	.51
Stroke	184 (2.1)	0.70	159 (1.8)	0.60	1.16 (0.94–1.44)	.16
ALI	43 (0.5)	0.16	38 (0.4)	0.14	1.13 (0.73–1.75)	.58
VTE	53 (0.6)	0.20	52 (0.6)	0.20	1.01 (0.69–1.49)	.95
Cancer						
All new cancers	429 (4.9)	1.65	435 (4.9)	1.77	0.99 (0.87–1.13)	.87
GI	86 (1.0)	0.33	83 (0.9)	0.31	1.04 (0.77–1.40)	.81
Lung	73 (0.8)	0.28	77 (0.9)	0.29	0.95 (0.69–1.31)	.75
Prostate	65 (0.7)	0.25	73 (0.8)	0.28	0.89 (0.64–1.24)	.50
Skin	73 (0.8)	0.28	70 (0.8)	0.26	1.05 (0.75–1.45)	.79
Breast	9 (0.1)	0.034	18 (0.2)	0.068	0.50 (0.22–1.11)	.08
Hospitalizations						
All	3074 (35.0)	14.51	3000 (34.1)	13.96	1.04 (0.99–1.09)	.14
Cardiovascular	1721 (19.6)	7.26	1644 (18.7)	6.86	1.06 (0.99–1.13)	.10
Non-cardiovascular	1898 (21.6)	8.13	1901 (21.6)	8.10	1.00 (0.94–1.07)	.92

ALI, acute limb ischemia; CHD, coronary heart disease; CI, confidence interval; GI, gastrointestinal; MI, myocardial infarction; od, once daily; VTE, venous thromboembolism.

*Defined by the cardiovascular outcomes related to aspirin rivaroxaban arms.

Outcome event penyakit cardiovascular (MI, Stoke, ALI, VTE) hingga cancer tidak ada perbedaan signifikan dengan placebo

Pantoprazole 40mg 1x1 selama 3 tahun aman dan tidak menimbulkan efek serius pada seseorang dengan penyakit arteri perifer yang mengkonsumsi Rivaroxaban dan Aspirin

ORIGINAL ARTICLE

Background/Aims: Concerns that proton pump inhibitors (PPIs) diminish the efficacy of clopidogrel could hamper the appropriate prescription of PPIs. We evaluated the influence of pantoprazole on the antiplatelet effect of clopidogrel compared with ranitidine, which is regarded as safe, after stratification of the population according to the presence of a cytochrome (CYP) 2C19 polymorphism in Korea. **Methods:** Forty patients who underwent dual antiplatelet therapy were randomized to receive pantoprazole (n=20) or ranitidine (n=20). Platelet aggregation was evaluated by impedance aggregometry at baseline (D0) and 8 days after acid-lowering treatments (D9). CYP2C19 was genotyped by polymerase chain reaction–restriction fragment length polymorphism.

Results: After co-treatment, the percentage of clopidogrel low-response was 11.1% (2/18) in the pantoprazole group and 10.5% (2/19) in the ranitidine group (p=0.954). The impedance values with adenosine diphosphate stimulus after acid-lowering treatments did not significantly differ between the two groups. In a multiple regression analysis, only ST-elevation myocardial infarction was marginally associated with a reduced antiplatelet effect (odds ratio, 12.07; 95% confidence interval, 0.84 to 173.78). However, pantoprazole use did not affect the antiplatelet effect after correction for the CYP2C19 polymorphism. **Conclusions:** This study showed that pantoprazole does not increase platelet aggregation in patients receiving dual antiplatelet therapy (ClinicalTrials.gov number: NCT02733640). (Gut Liver 2017;11:504-511)

Pantoprazole Does Not Reduce the Antiplatelet Effect of Clopidogrel: A Randomized Controlled Trial in Korea

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Table 2. Change in Impedance and Concentration of the Active Clopidogrel Metabolite in Each Acid-Lowering Treatment Group

	Ranitidine (n=20)			Pantoprazole (n=20)			p-value [†]	p-value [‡]
	Pretreatment (D0)	Posttreatment (D9)	p-value*	Pretreatment (D0)	Posttreatment (D9)	p-value*		
Impedance (Ω)	1.00 (0.25–3.50)	1.00 (0.00–2.00)	0.316	1.50 (0.25–4.00)	2.00 (0.25–4.00)	0.775	0.547	0.108
Clopidogrel active metabolite	0.36 (0.19–1.47)	0.67 (0.01–4.14)	0.327	0.18 (0.01–2.62)	0.13 (0.01–2.26)	0.575	0.519	0.370

Data are presented as median (interquartile).

*Analyses between pretreatment and posttreatment using the Wilcoxon rank-sum test in each group; [†]Analyses between pretreatment groups using the Mann-Whitney test; [‡]Analyses between posttreatment groups using the Mann-Whitney test.

Pantoprazole tidak meningkatkan agregasi platelet pada pasien yang menerima terapi dual antiplatelet

Abstract: Recent attention has been drawn to a potential drug–drug interaction observed between clopidogrel and proton pump inhibitors (PPIs). However, this potential interaction may not be a class effect of PPIs. We investigated if pantoprazole, which has a different metabolism than omeprazole, diminishes the effectiveness of clopidogrel. Our study included 336 patients (mean age 64.6 years; 106 women) 48 hours after percutaneous coronary stent implantation with a loading dose of 600 mg clopidogrel hydrogensulfate and 500 mg aspirin, followed by 75 mg clopidogrel and 100 mg aspirin daily. Whereas 188 patients (59 women) were not given any PPI comedication, 122 patients received pantoprazole and 26 either omeprazole or esomeprazole. The platelet aggregation followed by impedance aggregometry (in Ohm) was induced by 5 μ mol/L adenosine diphosphate. The percentage of clopidogrel low-response (CLR) was similar between the non-PPI group [2.75 Ohm (confidence interval, CI: 2.25–3.26); 21.9% CLR] and the pantoprazole group [2.33 Ohm (CI: 1.79–2.87); 16.4% CLR] but higher in patients treated with omeprazole/esomeprazole (3.00 Ohm (CI: 1.49–4.51); 30.8% CLR). Multivariate regression analysis reveals that the risk of CLR in the pantoprazole comedication group was not increased compared with the group without any PPI [odds ratio 0.59 (CI: 0.31–1.13) 0.11]. Our data suggest that pantoprazole does not diminish the antiplatelet effectiveness of clopidogrel early after coronary stenting. Therefore, the use of pantoprazole seems preferable in patients treated with clopidogrel when a concomitant medication with a PPI is indicated.

Pantoprazole Does Not Influence the Antiplatelet Effect of Clopidogrel—A Whole Blood Aggregometry Study After Coronary Stenting

Horst Neubauer, MD,* Andreas Engelhardt, MD,* Jan C. Krüger, MS,* Sebastian Lask, MD,* Jan Börgel, MD,* Andreas Mügge, PhD,* and Heinz G. Endres, MD†

Pantoprazole dapat direkomendasikan sebagai pilihan PPI yang lebih aman bagi pasien yang membutuhkan terapi clopidogrel dibandingkan PPI lainnya

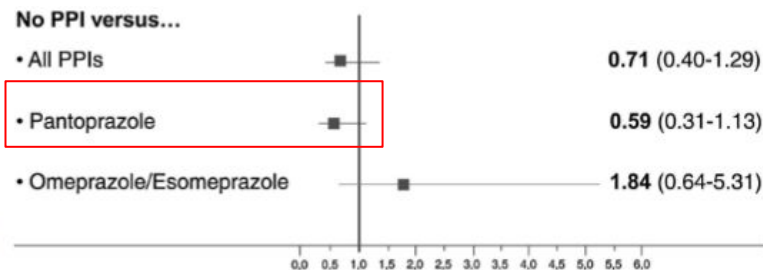


FIGURE 1. Forest plot of risk ratios (OR) of clopidogrel low-response (CLR) in different PPI groups.

Antiplatelet Therapy

Comparison of Omeprazole and Pantoprazole Influence on a High 150-mg Clopidogrel Maintenance Dose

The PACA (Proton Pump Inhibitors And Clopidogrel Association) Prospective Randomized Study

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Bénédicte Gaborit, MD,†‡ Laurent Bali, MD,* Olivier Brissy, MD,*
Pierre-Emmanuel Morange, MD, PhD,†‡ Marie-Christine Alessi, MD, PhD,†‡
Jean-Louis Bonnet, MD*

Objectives

This study sought to compare the effect of 2 proton pump inhibitors (PPIs) on platelet response to clopidogrel after coronary stenting for non-ST-segment elevation acute coronary syndrome (NSTEMI ACS).

Background

Use of omeprazole has been reported to significantly decrease the clopidogrel antiplatelet effect because of cytochrome P450 interaction. Because all PPIs are metabolized by CYP2C19, but to a varying degree, we hypothesized that the reported negative omeprazole-clopidogrel drug interaction may not be caused by a class effect.

Methods

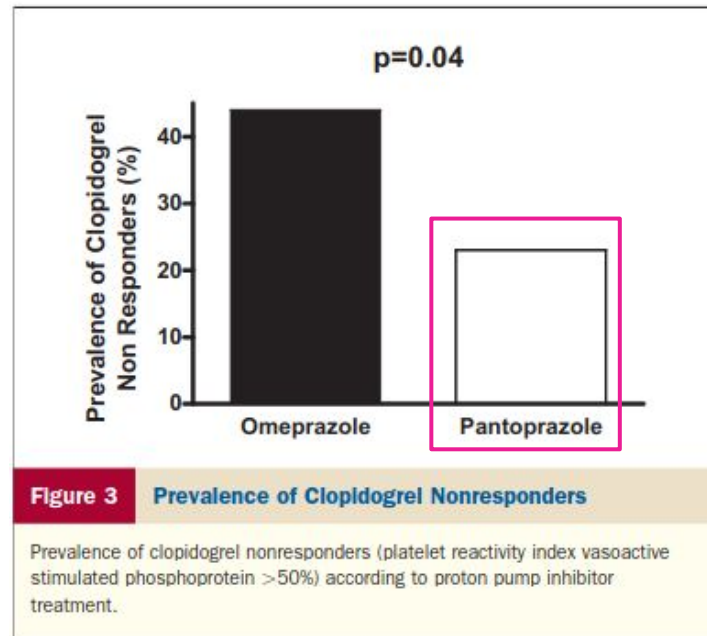
A total of 104 patients undergoing coronary stenting for NSTEMI ACS were prospectively included and randomized to omeprazole or pantoprazole 20 mg. They received at discharge 75-mg aspirin and 150-mg clopidogrel. Platelet reactivity index (PRI) vasoactive stimulated phosphoprotein (VASP) was used to assess clopidogrel response and adenosine diphosphate (ADP)-induced aggregation for platelet reactivity (ADP-Ag).

Results

After 1 month, patients receiving pantoprazole had a significantly better platelet response to clopidogrel as assessed with the PRI VASP: $36 \pm 20\%$ versus $48 \pm 17\%$ ($p = 0.007$). We identified more clopidogrel nonresponders in the omeprazole group than in the pantoprazole group: 44% versus 23% ($p = 0.04$), odds ratio: 2.6 (95% confidence interval: 1.2 to 6.2). Conversely, we did not observe any significant difference in platelet reactivity with ADP-Ag between the omeprazole and pantoprazole groups: $52 \pm 15\%$ and $50 \pm 18\%$, respectively ($p = 0.29$).

Conclusions

The present findings suggest the preferential use of pantoprazole compared with omeprazole in patients receiving clopidogrel to avoid any potential negative interaction with CYP2C19. (J Am Coll Cardiol 2009;54:1149-53) © 2009 by the American College of Cardiology Foundation

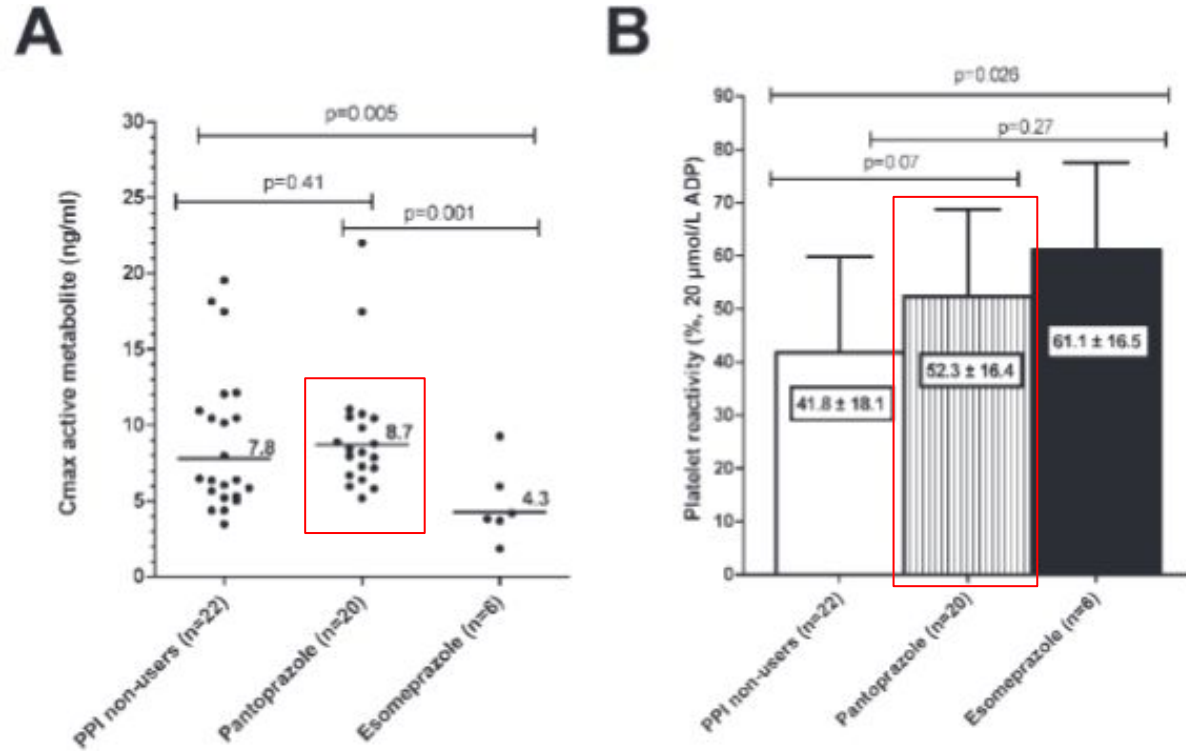


Setelah 1 bulan, kelompok pantoprazole menunjukkan respon antiplatelet klopidogrel lebih baik dibanding kelompok omeprazole.

Omeprazole menurunkan efek antiplatelet klopidogrel dengan dosis klopidogrel ditingkatkan, sedangkan pantoprazole tidak menurunkan efektivitas klopidogrel

Efek Pantoprazole vs Esomeprazol pada Clopidogrel

Pantoprazole 40mg lebih aman digunakan bersama clopidogrel dibandingkan dengan esomeprazole, terutama pada pasien yang membutuhkan terapi PPI bersamaan dengan clopidogrel



Summary

Pada pasien CAD, AF, VTE, PAD, atau pasca-stroke iskemik yang mengonsumsi antiplatelet/antikoagulan

PPI dapat menurunkan efektivitas dari kerja CPG

Pantoprazole merupakan PPI yang paling sedikit menghambat CYP2C19, sehingga interaksi negatif dengan clopidogrel nyaris tidak terjadi » efek antiplatelet tetap optimal.

Pantoprazole menurunkan risiko perdarahan gastrointestinal atas tanpa meningkatkan kejadian kardiovaskular mayor dibandingkan PPI lainnya